

A manifestly positive functional system for the cylindric-partition generating functions at $d = 7$ (modulus 10) Seed 3, Layer 4, Round 2 — Template B at the first unproved level

Loop experiment, 2026-07-04

Abstract

We construct a Corteel–Welsh-style manifestly positive y -functional system (the **Eq:Fun** analogue) for all twelve C_3 -orbits of profiles at $d = 7$ (modulus 10), the smallest level at which no positivity result of any kind was previously known in the literature (internal Layer-2 propagation covered the zero-containing orbits; see §1). The system (i) has all coefficients in $\mathbb{N}[y, q]$ and all argument shifts yq^s with $s \geq 1$, (ii) determines all $G_c(y, q)$ uniquely from $G_c(0, q) = 1$ by a q -adic contraction, and (iii) reproduces the raw inclusion–exclusion (Corteel–Welsh) solution exactly. Every row is *derived* — an explicit finite chain of substitutions of proved identities into proved identities — not fitted. Consequence: $G_c(y, q) \in \mathbb{N}[[y, q]]$ for every profile c at $d = 7$, closing the all-positive core left open by the Layer-2 Propagation Theorem at this level. Extraction of $Q_{n,c} \geq 0$ (bounded forms and absorption lemmas) is not attempted here.

1 Setup and conventions

Fix $d = 7$, so $\ell = \gcd(d, 3) = 1$ and the modulus is $d + 3 = 10$. Profiles are compositions $c = (c_0, c_1, c_2)$ with $c_0 + c_1 + c_2 = 7$; there are 36 of them in $K = (8 \cdot 9)/6 = 12$ orbits under the cyclic rotation $\sigma(c_0, c_1, c_2) = (c_1, c_2, c_0)$. All labels are the *true conjecture.tex* labels (synthesis-layer3 §4(iv)); at $d = 7$ the identity label map between the CW-note convention and the raw interlacing definition was pinned exactly by the Phase-1 engine (scratch log, Phase 1).

Let $F_c(y, q) = \sum_m F_{c,m}(q) y^m$ be the cylindric-partition generating function graded by largest part, and

$$G_c(y, q) = (yq; q)_\infty F_c(y, q) = \sum_{n \geq 0} g_c(n) y^n.$$

F_c , hence G_c , is invariant under rotation of the profile ($F_{\sigma c} = F_c$; the relabeling bijection of cylindric partitions, proved in *prove-seed2-layer2.tex*, Lemma 1). We may therefore normalise every profile to the lexicographically smallest rotation of its orbit; the twelve orbit representatives are

$$\begin{aligned} Z_1 &= (0, 0, 7), & Z_2 &= (0, 6, 1), & Z_3 &= (0, 1, 6), & Z_4 &= (0, 5, 2), \\ Z_5 &= (0, 2, 5), & Z_6 &= (0, 4, 3), & Z_7 &= (0, 3, 4), \\ C_1 &= (1, 1, 5), & C_2 &= (1, 4, 2), & C_3 &= (1, 2, 4), & C_4 &= (1, 3, 3), & C_5 &= (2, 2, 3), \end{aligned}$$

the Z 's zero-containing, the C 's the all-positive *core*.

The raw system. By Corteel–Welsh (Proposition **incex**, form **Gb**, of the companion note, applied at $k = 3$, $\ell = 10$),

$$G_c(y) = \sum_{\emptyset \neq J \subseteq I_c} (-1)^{|J|-1} (yq; q)_{|J|-1} G_{c(J)}(yq^{|J|}), \quad G_c(0, q) = G_c(y, 0) = 1, \quad (1)$$

where $I_c = \{i : c_i > 0\}$ and $c(J)$ subtracts 1 at each $i \in J$ with $i - 1 \notin J$ and adds 1 at each $i \notin J$ with $i - 1 \in J$ (indices mod 3). Each instance of (1) is a proved identity in $\mathbb{Z}[[y, q]]$.

Rows and exact substitution. A row for a target orbit t is an identity

$$G_t(y) = \sum_{(r,s)} m_{r,s}(y, q) G_r(yq^s), \quad m_{r,s} \in \mathbb{Z}[y, q],$$

with r ranging over orbit representatives. Given a proved row for r , one may substitute it for the factor $G_r(yq^s)$ of any term (evaluating the row at $y \mapsto yq^s$) and collect; the result is again a proved row. All derivations below are finite chains of such substitutions starting from instances of (1); *no step is a fit*.

A row is *positive* if every $m_{r,s} \in \mathbb{N}[y, q]$ and every shift satisfies $s \geq 1$. Setting $y = 0$ in a positive row forces exactly one term with unit $y^0 q^0$ -coefficient (the *head*).

2 The zero-containing rows

Lemma 1 (R-rows at $d = 7$). *The following positive rows hold; the Family A/B constructions of the Layer-2 Substitution Lemma (prove-seed6-layer2, §2), run mechanically at $d = 7$ for every composition in each orbit, produce exactly these rows (Families A and B coincide orbit-level):*

$$\begin{aligned} G_{Z_1}(y) &= G_{Z_2}(yq), \\ G_{Z_2}(y) &= G_{Z_3}(yq) + yq G_{C_1}(yq^2) + yq^2 G_{C_2}(yq^3) + yq^3 G_{C_4}(yq^4) + yq^4 G_{C_3}(yq^5) \\ &\quad + yq^5 G_{C_1}(yq^6) + yq^6 G_{Z_2}(yq^7), \\ G_{Z_3}(y) &= G_{C_1}(yq) + yq G_{Z_2}(yq^2), \\ G_{Z_4}(y) &= G_{C_1}(yq) + yq G_{C_2}(yq^2) + yq^2 G_{C_4}(yq^3) + yq^3 G_{C_3}(yq^4) + yq^4 G_{C_1}(yq^5) + yq^5 G_{Z_2}(yq^6), \\ G_{Z_5}(y) &= G_{C_3}(yq) + yq G_{C_1}(yq^2) + yq^2 G_{Z_2}(yq^3), \\ G_{Z_6}(y) &= G_{C_2}(yq) + yq G_{C_4}(yq^2) + yq^2 G_{C_3}(yq^3) + yq^3 G_{C_1}(yq^4) + yq^4 G_{Z_2}(yq^5), \\ G_{Z_7}(y) &= G_{C_4}(yq) + yq G_{C_3}(yq^2) + yq^2 G_{C_1}(yq^3) + yq^3 G_{Z_2}(yq^4). \end{aligned}$$

Proof. Family A: for $c = (a, 0, b)$ with $b \geq 1$, substitute the (already derived) row of $(a+1, 0, b-1)$ into the single term $G_{(a+1,0,b-1)}(yq)$ of (1); its head $G_{(a,1,b-1)}(yq^2)$ cancels the pair term $-(1 - yq)G_{(a,1,b-1)}(yq^2)$ up to $+yq G_{(a,1,b-1)}(yq^2)$; induction terminates at $(7, 0, 0)$, whose raw row is the single positive term $G_{(6,1,0)}(yq)$. Family B: for $c = (a, b, 0)$, substitute the Family-A row of $(b+1, 0, a-1) \sim (a-1, b+1, 0)$ (for $a = 1$, the raw $|I| = 1$ row of $(0, b+1, 0)$). Every step is the Substitution Lemma verbatim; the mechanical replay and orbit-normalisation are in `seed3_R2L4_construct.py` (`derive_zero_rows`), each output row verified against the raw-system solution for $n \leq 8$ to q^{200} . $\square$

3 The core rows

Lemma 2 (core rows at $d = 7$). *The following positive rows hold:*

$$G_{C_1}(y) = G_{C_2}(yq) + yq G_{Z_4}(yq^2) + yq G_{C_3}(yq^2) + (yq^2 + y^2 q^3) G_{C_1}(yq^3) + (yq^3 + y^2 q^4) G_{Z_2}(yq^4),$$

$$G_{C_3}(y) = G_{C_5}(yq) + yq G_{C_4}(yq^2) + yq G_{C_2}(yq^2) + yq^2 G_{Z_4}(yq^3) + (yq^2 + y^2 q^3) G_{C_3}(yq^3) \\ + (yq^3 + y^2 q^4 + y^2 q^5) G_{C_1}(yq^4) + (yq^4 + y^2 q^5 + y^2 q^6) G_{Z_2}(yq^5),$$

$$G_{C_4}(y) = G_{C_5}(yq) + yq G_{C_2}(yq^2) + yq G_{C_5}(yq^2) + (yq^2 + y^2 q^3) G_{C_4}(yq^3) + yq^2 G_{C_2}(yq^3) + yq^3 G_{Z_4}(yq^4) \\ + (yq^3 + y^2 q^4 + y^2 q^5) G_{C_3}(yq^4) + (yq^4 + y^2 q^5 + y^2 q^6 + y^2 q^7) G_{C_1}(yq^5) \\ + (yq^5 + y^2 q^6 + y^2 q^7 + y^2 q^8) G_{Z_2}(yq^6),$$

$$G_{C_2}(y) = G_{C_4}(yq) + yq G_{Z_6}(yq^2) + yq G_{C_5}(yq^2) + yq^2 G_{C_4}(yq^3) + (yq^2 + y^2 q^3) G_{C_2}(yq^3) \\ + (yq^3 + y^2 q^4) G_{Z_4}(yq^4) + (yq^3 + y^2 q^4 + y^2 q^5) G_{C_3}(yq^4) \\ + (yq^4 + y^2 q^5 + y^2 q^6 + y^2 q^7 + y^3 q^8) G_{C_1}(yq^5) + (yq^5 + y^2 q^6 + y^2 q^7 + y^2 q^8 + y^3 q^9) G_{Z_2}(yq^6),$$

$$G_{C_5}(y) = G_{C_5}(yq) + yq G_{C_4}(yq^2) + yq G_{C_5}(yq^2) + yq^2 G_{Z_6}(yq^3) + yq^2 G_{C_2}(yq^3) + (yq^2 + y^2 q^3) G_{C_5}(yq^3) \\ + (yq^3 + y^2 q^4 + y^2 q^5) G_{C_4}(yq^4) + (yq^3 + y^2 q^4 + y^2 q^5) G_{C_2}(yq^4) + (yq^4 + y^2 q^5 + y^2 q^6) G_{Z_4}(yq^5) \\ + (yq^4 + y^2 q^5 + 2y^2 q^6 + y^2 q^7 + y^3 q^8) G_{C_3}(yq^5) \\ + (yq^5 + y^2 q^6 + 2y^2 q^7 + y^2 q^8 + y^2 q^9 + y^3 q^9 + y^3 q^{10} + y^3 q^{11}) G_{C_1}(yq^6) \\ + (yq^6 + y^2 q^7 + 2y^2 q^8 + y^2 q^9 + y^2 q^{10} + y^3 q^{10} + y^3 q^{11} + y^3 q^{12}) G_{Z_2}(yq^7).$$

Proof. Each row is obtained from the raw row (1) of its target by the following explicit substitution chains (“ $X @ (r, s)$ ” means: substitute the row X into the term at (r, s) ; $\text{raw}[r]$ is (1) for r , $R[Z_i]$ the row of Lemma 1, $C[\cdot]$ a previously proved row of this lemma):

- C_1 : $\text{raw}[Z_2] @ (Z_2, 1)$; $R[Z_5] @ (Z_5, 1)$; $R[Z_3] @ (Z_3, 2)$.
- C_3 : $\text{raw}[C_1] @ (C_1, 1)$; $\text{raw}[Z_2] @ (Z_2, 2)$; $R[Z_7] @ (Z_7, 1)$; $R[Z_5] @ (Z_5, 2)$; $R[Z_3] @ (Z_3, 3)$.
- C_4 : $\text{raw}[C_3] @ (C_3, 1)$; $\text{raw}[C_1] @ (C_1, 2)$; $\text{raw}[Z_2] @ (Z_2, 3)$; $R[Z_6] @ (Z_6, 1)$; $R[Z_7] @ (Z_7, 2)$; $R[Z_5] @ (Z_5, 3)$; $R[Z_3] @ (Z_3, 4)$.
- C_2 : $\text{raw}[Z_4] @ (Z_4, 1)$; $\text{raw}[C_3] @ (C_3, 1)$; $\text{raw}[C_1] @ (C_1, 2)$; $\text{raw}[Z_2] @ (Z_2, 3)$; $R[Z_7] @ (Z_7, 2)$; $R[Z_5] @ (Z_5, 3)$; $R[Z_3] @ (Z_3, 4)$.
- C_5 : $C[C_4] @ (C_4, 1)$; $C[C_3] @ (C_3, 2)$; $C[C_2] @ (C_2, 1)$.

Every step is exact polynomial algebra on proved identities, so each endpoint is a proved identity; that the endpoints equal the displayed rows (and are positive) is a finite computation, replayed deterministically by `seed3_R2L4_system.py` (PATHS); in addition every row was verified against the raw-system solution for all $n \leq 8$ to q^{200} . The derivation order C_1, C_3, C_4, C_2, C_5 is acyclic: the first four use only raw rows and Lemma 1; C_5 uses C_2, C_3, C_4 . $\square$

Remark 3. *The mechanism differs from the depth-2 manipulation of the CW note at $d = 4$: raw rows are substituted down a chain of singles (core $\rightarrow \dots \rightarrow$ zero-containing), the negativity telescopes down the chain, and the accumulated zero-orbit terms are closed off with the R -rows, which absorb all residual negative monomials. The head graph is $C_1 \rightarrow C_2 \rightarrow C_4 \rightarrow C_5 \rightarrow C_5$ and $C_3 \rightarrow C_5$; the self-head of C_5 at shift 1 is exactly the shape of the CW $(2, 1, 1)$ row, giving the $(1 - q^n)$ -division step there.*

4 The theorem

Theorem 4 (positive system at $d = 7$). *The twelve rows of Lemmas 1 and 2 form a system with: (i) all coefficients in $\mathbb{N}[y, q]$ and all shifts ≥ 1 ; (ii) together with $g_c(0) = 1$, a unique solution $(G_c)_c$ in $\mathbb{Z}[[y, q]]^{12}$; and (iii) that solution is the cylindric-partition family: it coincides with the solution of the raw system (1).*

Proof. (i) is manifest from the displays. (iii) holds because each row was derived from (1) by exact substitutions, so the raw solution satisfies the system; uniqueness (ii) then gives coincidence. For (ii), extract $[y^n]$ of the row of t :

$$g_t(n) = \sum_{(r,s)} \sum_{(j,a)} m_{r,s}[j, a] q^{a+s(n-j)} g_r(n-j),$$

where $m_{r,s} = \sum_{j,a} m_{r,s}[j, a] y^j q^a$. Terms with $j \geq 1$ refer to levels $n-j < n$. The level- n subsystem is $g(n) = M_n g(n) + b_n$ with b_n determined by levels $< n$ and $(M_n)_{t,r} = \sum_{s,a} m_{r,s}[0, a] q^{a+sn}$, every entry of q -adic valuation $\geq n \geq 1$. Hence $(I - M_n)^{-1} = \sum_{k \geq 0} M_n^k$ converges q -adically and $g(n) = (I - M_n)^{-1} b_n$ is unique. Induction on n from $g(0) = 1$. $\square$

Corollary 5 (bivariate positivity at $d = 7$). *For every profile c with $c_0 + c_1 + c_2 = 7$, all coefficients of $G_c(y, q)$ (equivalently, all $g_c(n) \in \mathbb{N}[[q]]$) are nonnegative. In particular the all-positive core $\{C_1, \dots, C_5\}$, unreachable by the Layer-2 Propagation Theorem, is settled at the G -level.*

Proof. In the proof of Theorem 4, M_n and b_n have nonnegative coefficients whenever the lower levels do, and $(I - M_n)^{-1} = \sum_k M_n^k$ preserves nonnegativity; induction on n with $g(0) = 1 \geq 0$. $\square$

Remark 6 (what this does and does not give). *The corollary is the $d = 7$ analogue of the positivity that Warnaar and Corteel–Dousse–Uncu establish en route at $d = 5$, and fulfils requirements (i)–(iii) of the Layer-3 Mission 3 brief (Template B, construction step). It does not by itself yield Warnaar’s conjecture $Q_{n,c} = (q; q)_n g_c(n) \geq 0$ at $d = 7$: that requires solving the system in closed form (Thm. G -style bounded multisums) and absorption lemmas for the resulting $(1 - q^n)$ wall terms, which we leave to the next layer. The system here is the prerequisite for that step, and its uniqueness induction (Theorem 4(ii)) is exactly the induction that such closed forms must be checked against.*

Verification appendix

All computations are exact integer arithmetic (2026-07-04/scratch/scripts/): `seed3_R2L4_engine.py` (raw-system solver, validated in Phase 1 against brute-force cylindric enumeration, the seed8 target-first H -recursion port with identity label map, and $Q_n(1) = 11^n$); `seed3_R2L4_construct.py` (exact row algebra, R-row derivations); `seed3_R2L4_search.py` (substitution-chain search; log `seed3_R2L4_search.log`); `seed3_R2L4_system.py` (deterministic replay of the five chains, assembly, and the standalone fixed-point solve of the positive system, which reproduces the raw solution for all 12 orbits, $n \leq 8$, to q^{200} ; log `seed3_R2L4_system.log`; system pickled in `seed3_R2L4_system.d7.pk1`).